Using shift-share instrument to study the impact of immigration on gubernatorial election outcomes

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Abstract

Gubernatorial election data provide a useful setting to examine whether immigrant voters are more likely to support the Democratic Party due to identity-based motivations. This paper empirically examines the causal effect of immigration on gubernatorial election outcomes using state-level panel data. A key empirical challenge is the endogeneity of immigration, as migration patterns are jointly determined by economic conditions and political environments.

I begin by assessing a commonly used instrument based on aggregate federal immigration flows and show that it suffers from weak identification. To overcome these limitations, I construct a shift-share instrumental variable that exploits historical settlement patterns and variation in national inflows across origin countries.

The empirical results indicate that the shift-share approach provides stronger identification and more robust estimates. More broadly, this study demonstrates a structured approach to causal inference, including diagnosing endogeneity, constructing theoretically grounded instruments, and systematically evaluating alternative empirical strategies.

1. Introduction

In recent years, the political polarization in the United States has become increasingly pronounced, driven in part by the impacts of globalization and waves of immigration. As Autor et al. (2020) point out, between 2004 and 2015, the number of political moderates declined while the number of individuals with extreme political views increased. This polarization also exhibited clear racial differences: non-Hispanic white voters increasingly supported the Republican Party, whereas Hispanic and non-white voters increasingly aligned with the Democratic Party. The influence of racial identity on voting behavior in presidential elections has been extensively documented in the literature. However, do immigrant voters also strongly support the Democratic Party in gubernatorial elections?

Although state governments cannot issue visas, permanent residence, or citizenship to immigrants or make international trade policies as the federal government does, they do have some capacity to influence immigrants' living conditions, rights, and access to public services within their borders. Therefore, if an anti-immigrant candidate is likely to win a gubernatorial election, immigrants who are sensitive to policy changes may leave the current "hostile" state and move to more immigrant-friendly states, where they tend to support Democratic candidates.

The immigrant ratio (the independent variable) will thus be correlated with the unobservable "state color" captured in the error term, leading to endogeneity.

This endogeneity issue can be addressed using an instrumental variables (IV) approach. A valid instrument must satisfy both the exogeneity and relevance conditions. Intuitively, federal-level immigration appears to be a plausible instrument; however, in the next section, I will explain why this variable is unlikely to be a valid instrument (as will be reflected in the regression results reported in Table 2 of Section 4), and propose that a shift-share instrument is more suitable for addressing the endogeneity problem.

2. Modeling analysis: Why is SSIV a better instrumental variable than federal immigration inflows?

The purpose of this paper is to get the average treatment effect of β_1 in equation (1). In equation (1), i denotes state, t denotes year, $Democrat_{it}$ is the proportion of votes received by the Democratic candidate in a gubernatorial election, $immigration\ ratio_{it}$ is the ratio of naturalized immigrants to the total population, ϵ_{it} is the error term.

$$Democrat_{it} = \beta_0 + \beta_1 immigration\ ratio_{it} + control\ variables_{it} + \epsilon_{it} \qquad (1)$$

However, immigrants who are sensitive to immigration-related policies are more likely to leave states that are unfriendly toward immigrants and choose to reside in states that are more immigrant-friendly. Because of this sensitivity, they are also more likely to vote for Democrats, who are generally more supportive of immigrants than Republicans. In contrast, immigrants who are not concerned with immigration-related policies have no particular preference over which state they live in, and their support for Democrats is correspondingly weaker. In other words, the error term ϵ_{it} in equation (1) can be written as

$$\epsilon_{it} = policy\ friendliness_{it} + \varepsilon_{it} \qquad (2)$$

Where ε_{it} is the real random error uncorrelated with any other variables. $state\ friendliness_{it}$ is correlated with $immigration\ ratio_{it}$, causing the endogeneity problem. This endogeneity problem can be partially treated by a state fixed effect; however, the state fixed effect may not perfectly solve the endogeneity problem as the friendliness of policies can change over time. Different politicians and different governments make different policies, and the "color" of a state won't remain stable. For example, in many strongly Democratic states (such as New York and California), Republican candidates won approximately 25% to 50% of gubernatorial elections between 1978 and 2022, while Democratic candidates also secured a considerable share of elections in strongly Republican states.

An instrumental variable approach can be applied to treat the endogeneity problem. An intuitive instrument is the federal immigrant ratio, because the immigration into the U.S. appears to cause the increase of immigrant numbers in each state (relevance), and because the total number of immigrants into the U.S. can hardly be influenced by the friendliness of any single state (exogeneity). However, a premise for "relevance" is that immigrants are evenly distributed across states; if this premise does not hold in reality, the relevance of the instrumental variable may be weak, leading to a weak instrument problem. In fact, immigrants from different original countries prefer to settle at different states. Figure 1 shows the shares of immigrants from Canada, China, and Mexico in the population of each U.S. state. States shaded closer to red indicate a higher share of naturalized immigrants from that country in the state's citizen population, while states shaded closer to green indicate a lower share. People from the three countries show clear differences in their preferred destination states for immigration. Canadians prefer states close to Ontario, British Columbia, and Alberta; Chinese prefer New York and California; Mexicans prefer states that share borders with Mexico. Assume that nearly all immigrants to the United States come from Canada, China, and Mexico. In certain years, suppose there is a surge in immigrants from Mexico, while there are no additional immigrants from Canada or China. In this case, total immigration to the United States would increase, but this growth would be concentrated mainly in states such as California, Nevada, New Mexico, Arizona, and Texas. These states would experience a sharp rise in immigrant populations, while immigrant numbers in other states would remain largely unchanged. Therefore, national-level immigration would appear less correlated with state-level immigration.

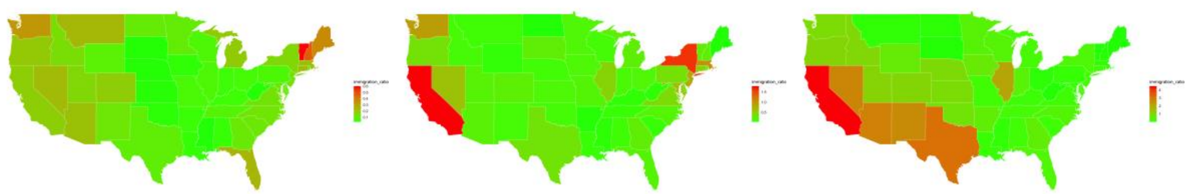

Figure 1 the preferred immigration destinations of Canadians, Chinese, and Mexicans

To better address the endogeneity problem, an SSIV approach can be applied. Assume that nearly all immigrants to the U.S. come from country A and country B, the value of SSIV is constructed following Borusyak, Hull, and Jaravel (2025):

$$\begin{aligned} SSIV_{it} = {} & A\ immigrant\ ratio_{i,baseyear} \times (1 + federal\ A\ increase\ rate_t) \\ & + B\ immigrant\ ratio_{i,baseyear} \times (1 + federal\ B\ increase\ rate_t) \end{aligned} \tag{3}$$

Here, $A\ immigrant\ ratio_{i,baseyear}$ denotes the proportion of naturalized A-country immigrants in state i in the base year, reflecting the preference of immigrants from country A. $federal\ A\ increase\ rate_t$ represents the national-level growth rate of naturalized A-country immigrants from the base year to year t. For example, if the base year is 2000 and there were 1 million A-country immigrants in the U.S. that year, and by 2010 that number grew to 2 million, then $federal\ A\ increase\ rate_{2010} = 1$.

The value of SSIV can be decomposed into the share part (e.g., $A\ immigrant\ ratio_{i,baseyear}$) and the shift part (e.g., $(1 + federal\ A\ increase\ rate_t)$). The shift part is plausibly exogeneous to ϵ_{it} in equation (1), because the number of federal immigrants is unlikely to be influenced by the outcomes of any gubernatorial elections. Therefore, even if the share part is endogenous (which may not be true), the SSIV is still plausibly exogeneous (Borusyak, Hull, and Jaravel, 2025).

A potential concern is that SSIV may not be sufficiently correlated with the actual state-level immigration data because immigrants' destination preference may substantially change over time. I will examine the correlation between SSIV and state-level immigration in weak instrument test to address this concern.

3. Empirical Design

3.1 The regression function

This paper's regression function is

$$\begin{aligned} Democrat_{it} = \ & \beta_0 + \beta_1 immigrant\ ratio_{i,t-1} + \beta_2 incomegrow_{i,t-1} \\ & + \beta_3 \frac{stateincome_{i,t-1}}{federalincome_{t-1}} + \beta_4 morethantwo_{i,t} + year\ fixed\ effect \\ & + state\ fixed\ effect + \varepsilon_{it} \end{aligned} \tag{4}$$

$Democrat_{it}$ denote the vote share received by the Democratic candidate in the gubernatorial election in state i and year t.

$immigrant\ ratio_{i,t-1}$represents the proportion of immigrants in the total population of state i in year t–1.

$incomegrow_{i,t-1}$refers to the per capita personal income of state *i* in year *t–1*,

$federalincome_{t-1}$indicates the national-level per capita personal income in year *t–1*. Data for both variables are obtained from the Bureau of Economic Analysis (BEA).

$morethantwo_{i,t}$is a binary indicator equal to 1 if there were more than two gubernatorial candidates in state *i* during the election year *t*, and 0 otherwise.

3.2. The calculation of immigrant ratio

To construct the variable $immigrant\ ratio_{i,t-1}$, I used population microdata from the Integrated Public Use Microdata Series (IPUMS) for the years 2000 to 2023. The dataset includes information at the individual level on citizenship status, place of birth, state of residence, and person weights.

For citizenship, IPUMS codes are as follows: 0 indicates U.S. citizens, 1 refers to individuals born abroad to American parents, 2 denotes naturalized citizens, and values 3 or higher represent non-citizens. Accordingly, I first dropped all individuals with a citizenship value of 3 or greater. I then classified codes 0 and 1 as native-born citizens, and code 2 as immigrant citizens. Using this classification, I calculated the proportion of immigrants in each state by summing weighted observations.

3.2. The calculation of SSIV

Based on individuals' reported place of birth on IPUMS[i], I calculated the share (and number) of immigrants in each state in each year from each country[ii]. I set 2000 as the base year so that the share of immigrants from each country in each state in 2000 is the "share" part of the SSIV; to get the "shift" part of the SSIV, I aggregate the state-level number of immigrants for each origin country to obtain the corresponding national totals; using these totals, I then compute the immigrant shares in each year and the corresponding increase rate.

4. Empirical results

Table 1 shows the OLS regression results. The dependent variable is the vote share received by the Democratic candidate in the gubernatorial election; the core independent variable is the state-level immigrant ratio. In column (1), I include only the key independent variable in the regression specification; control variables and fixed effects are not included. Column (2) includes all the control variables; column (3) includes all the control variables and the fixed effects.

As shown in Table 1, the coefficient on Immigrant ratio is positive and statistically significant at the 1% level, regardless of whether control variables and fixed effects are included.

Table 1 The OLS regression

	Vote share received by the Democratic candidate		
Immigrant ratio	65.456***	56.173***	465.845***
	(17.753)	(20.981)	(118.813)
Income growth rate		17.054	38.224
		(15.817)	(29.053)
State income/federal income		2.300	-13.180
		(4.985)	(14.879)
More than two candidates		-6.050***	-6.283***
		(1.885)	(1.829)
Year fixed effect	No	No	Yes
State fixed effect	No	No	Yes
N	281	281	281

Standard errors in parentheses.

*** significant at 1% level

** 5% level * 10% level

Next, I applied a standard instrumental variable (IV) approach, using the federal-level immigrant ratio as the instrument. A two-stage least squares (2SLS) estimation is applied. Table 2 reports the first-stage results (using the instrument as the dependent variable).

In Table 2, the t-statistics of IV "Federal immigrant ratio" reports the results of weak instrument tests. In columns (1) and (2), the F-statistics (the square of the t-statistic) for the federal immigrant ratio are slightly above the conventional threshold of 10. However, the column (3) shows a negative and insignificant partial correlation. The inclusion of time- and state- fixed effects in the column (3) alters the conditional correlation structure between national immigration and state-level immigration. This suggests that changes in state-level immigration are more driven by state-specific factors and common time trends, leaving limited independent variation in national immigration to explain state-level outcomes. Therefore, federal immigration is not a reliable instrument, which also confirms the concerns I raised in the modeling analysis section; there is no need to present the second-stage results of the IV (federal immigrant ratio) approach.

Table 2 The first-stage results for IV approach			
	State immigration ratio		
Federal immigrant ratio	0.751***	0.789***	-9.245
	(0.214)	(0.188)	(6.606)
Income growth rate		-0.002	-0.002
		(0.046)	(0.017)
State income/federal income		0.132***	-0.035***
		(0.012)	(0.008)
More than two candidates		-0.005	-0.001
		(0.005)	(0.001)
Year fixed effect	No	No	Yes
State fixed effect	No	No	Yes
N	281	281	281
t-statistics of IV "Federal immigrant ratio"	3.506	4.202	-1.399

Since the federal immigration is not a good instrument, I try to apply SSIV to address the endogeneity concern. A two-stage least squares (2SLS) estimation is applied. Table 3 reports the first-stage results (using the instrument as the dependent variable), and Table 4 reports the second-stage results (using the fitted value as the core independent variable).

In all the three columns of Table 3, the SSIV coefficients are positive, and the t-statistics are strikingly large, suggesting that SSIV is very unlikely to be a weak instrument. In all the columns of Table 4, the coefficients are uniformly positive and statistically significant at least at the 5% level. Therefore, the conclusion is robust.

Table 3 The first results for SSIV approach

	State immigration ratio		
SSIV	0.708***	0.633***	0.248***
	(0.029)	(0.029)	(0.016)
Income growth rate		-0.061	-0.002
		(0.028)	(0.012)
State income/federal income		0.063***	-0.019***
		(0.008)	(0.006)
More than two candidates		0.001	0.001
		(0.003)	(0.001)
Year fixed effect	No	No	Yes
State fixed effect	No	No	Yes
N	281	281	281
t-statistics of IV "SSIV"	24.075	21.633	15.196

Table 4 The second-stage results for SSIV approach			
	vote share received by the Democratic candidate		
Fitted value	76.714***	62.764**	705.505***
	(21.646)	(26.527)	(162.756)
Income growth rate		16.685	38.733
		(15.887)	(28.835)
State income/federal income		1.439	-4.888
		(5.426)	(15.268)
More than two candidates		-6.000***	-6.153***
		(1.894)	(1.816)
Year fixed effect	No	No	Yes
State fixed effect	No	No	Yes
N	281	281	281

5. Discussion

Many variables exhibit a shift-share structure, where different shares may exert heterogeneous causal effects on the dependent variable and may also be correlated with the error term to varying degrees. The shift-share instrumental variable approach better captures these underlying causal patterns and therefore provides a more robust solution to endogeneity.

In this paper, immigrants from different countries exhibit heterogeneous destination preferences. If commonly used causal methods (including instrumental variables, regression discontinuity, and difference-in-differences) do not sufficiently capture this structure, the resulting conclusions may lack robustness. For example, federal immigration, the instrumental variable used in Table 2, fails to pass the weak instrument test. In contrast, the first-stage F-statistics (for weak instrument diagnostics) of the shift-share instrumental variable are at least 25 times larger than those of the federal immigration instrument.

Reference

Autor, D., Dorn, D., Hanson, G., Majlesi, K. (2020). Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure. American Economic Review, 110(10), 3139–3183.

Borusyak, K., Hull, P., Jaravel, X. (2025). A practical guide to shift-share instruments. Journal of Economic Perspectives, 39(1), 181-204.

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[i] Due to the severity of missing data across many origin countries, I focused on immigrants from the following 17 major origins: Africa, Canada, China, England, France, Germany, India, Italy, Japan, Korea, Mexico, the Philippines, Poland, Scotland, Thailand, and Vietnam; immigrants originating from these 17 countries constitute nearly the entirety of the U.S. immigrant population.

[ii] I excluded the District of Columbia, Hawaii, Alaska, and Tennessee from regressions due to the severe data missing problem.